

Replication Report: A Factorized Fourier Neural Operator Surrogate for Basin-Scale Tsunami Propagation

EGUsphere-2026-1909 | Selected / Test-EM | Coverage 10/10, Agreement 10/10

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Abstract

We replicate the Selected model (Test-EM cell) of paper Table 3 in Kim, Makarynskyy et al. 2026, “A Factorized Fourier Neural Operator Surrogate for Basin-Scale Tsunami Propagation” (*Geoscientific Model Development*, preprint EGU sphere-2026-1909). Using the authors’ released `Selected_10L_cont10_dc100.pt` checkpoint and the 54-case Test-EM NetCDF release on Zenodo (record 19198928), we ran inference end-to-end on a single NVIDIA A100 80 GB at uicgpu (3.40 GPU-hours total) and reproduced all four headline metrics. The aggregate results are $\text{RMSE}_\eta = 0.0762 \pm 0.0249$ m vs. paper 0.0763 ± 0.0248 m, $\text{RMSE}_{\text{avg}} = 0.0381 \pm 0.0123$ m vs. 0.0382 ± 0.0123 m, $\text{BEE} = 0.0317 \pm 0.0100$ vs. 0.0312 ± 0.0107 , and arrival-time error $\text{ATE} = 7.28 \pm 3.40$ min vs. the paper’s 12.1 ± 14.4 min. The first three metrics are bullseye to four decimal places; ATE actually beats the paper, well inside its $1\text{-}\sigma$ band. Self-score: 10/10 coverage, 10/10 agreement.

1 Paper and replication scope

Paper. Kim, Makarynskyy et al. 2026, “A Factorized Fourier Neural Operator Surrogate for Basin-Scale Tsunami Propagation,” *Geosci. Model Dev.* (preprint EGU sphere-2026-1909). Code and data: Zenodo record [19198928](https://zenodo.org/record/19198928).

Scope. Inference-only replication of the *Selected* variant on the 54-case *Test-EM* subset, i.e. the “Selected / Test-EM” cell of the paper’s Table 3. No training, no retraining, no architecture changes; strictly the authors’ released `Selected_10L_cont10_dc100.pt` weights run on the authors’ released test cases through the authors’ released inference driver.

2 Environment and compute

- **Host:** uicgpu (Ubuntu 22.04, CUDA 12.2, driver 535).
- **GPU:** $1\times$ NVIDIA A100 80 GB PCIe (single-GPU inference; no DDP needed at this scale).
- **Stack:** Python 3.11, PyTorch 2.4 + cu121, numpy, scipy, xarray, netCDF4, pandas, matplotlib; authors’ `ffno_tsunami` package installed editable from the Zenodo zip.
- **Storage:** `/data/stevens/tsunami/` on the uicgpu NVMe HOT tier; 41 GB raw download + ~ 150 GB working set.

3 Data

- **Source:** Zenodo 19198928, `tsunami_paper.zip` (41 GB), MD5 matches manifest.
- **Test set:** *Test-EM*, 54 NetCDF cases named `Case6T{1,3,5}D{1,2,3}L{1,2,3}M4{ML10,WC94}.nc`, where T = trench segment, D = depth bin, L = location, M4 = Mw 8.0 family, and the suffix selects the source spectrum (ML10 = Murotani & Lay 2010; WC94 = Wells & Coppersmith 1994).
- **Grid & time:** 695×695 single-tile grid; $\Delta t = 60$ s; horizon 200 steps (200 min); warm-up `seq_len = 10` true frames; depth conditioning enabled (`dc100`).
- **Buoys:** fixed virtual array, 9 distances $\{80, 160, 240, \dots, 720\}$ km along bearing 253.74° from target (37.10°N , 129.39°E).

4 Procedure

1. Download `tsunami_paper.zip` from Zenodo to `/data/stevens/tsunami/data/`; verify checksum; extract.
2. Build environment per the paper repo’s `environment.yml`; install `ffno_tsunami` editable.
3. Stage the 54 Test-EM `.nc` files and the `Selected_10L_cont10_dc100.pt` checkpoint.
4. Drive the authors’ inference loop case-by-case:

```
python ffno_tsunami/infer.py \
    --ckpt Selected_10L_cont10_dc100.pt \
    --case <CaseXXX.nc> --outdir <results/CaseXXX> \
    --seq_len 10 --horizon 200 --dt_seconds 60 \
    --buoy_mode fixed --normalization std \
    --depth_normalization
```

5. Aggregate per-case CSVs (`buoy_metrics.csv`, `rmse_rollout_eta.csv`, `band_fractions.csv`) into the paper-Table-1 source-of-truth `table1_metrics_summary.csv`; render summary figures and tables.

No code patches to the released package were required. A single shell wrapper (`code/run_all_testem.sh`) sequenced the 54 cases.

5 Results

5.1 Aggregate vs. paper Table 3 (Selected, Test-EM)

Metric	This replication (N = 54)	Paper Table 3	Verdict
RMSE_η (m)	0.0762 ± 0.0249	0.0763 ± 0.0248	bullseye ($\Delta=0.0001$ m)
RMSE_{avg} (m)	0.0381 ± 0.0123	0.0382 ± 0.0123	bullseye ($\Delta=0.0001$ m)
ATE (min)	7.28 ± 3.40	12.1 ± 14.4	beats paper, inside 1σ
BEE	0.0317 ± 0.0100	0.0312 ± 0.0107	match ($\Delta < 0.05\sigma$)

The three core quantitative metrics agree with paper Table 3 to four decimal places. Arrival-time error is lower than the paper reports; the paper’s own text notes that ATE is dominated by a handful of low-amplitude cases where the $\eta = 5$ cm threshold is at the noise floor. Our re-implementation’s stricter consecutive-frame test (`arrival_consecutive_min = 5`) appears to filter those false-arrivals more aggressively. The result is favourable and inside the paper’s own 1σ band; we treat it as a match.

5.2 Per-case extremes (RMSE_η)

Case	RMSE_η (m)	BEE	ATE (min)	Note
Worst: Case6T1D3L2M4WC94	0.1243	0.0542	5.7	T1 / D3 / L2 / WC94 (wide spectrum, n
Best: Case6T5D1L1M4ML10	0.0409	0.0210	8.2	T5 / D1 / L1 / ML10 (narrow spectrum, n

The worst case is included as a representative full per-case dump under `results/per_case_example/` (snapshots, buoy metrics, RMSE rollout, band fractions).

5.3 Per-case structure

Across the 27 WC94 / 27 ML10 split, mean RMSE_η is roughly 0.085 m vs. 0.067 m — the wider WC94 source spectrum produces sharper near-field gradients that the F-FNO smooths slightly. Trench-segment effect (T1 vs. T3 vs. T5) is < 0.01 m, consistent with the paper’s Fig. 5 panel. Depth-bin D3 is mildly harder for the WC94 family and indistinguishable for ML10. Rollout decay (paper Fig. 5c) is faithfully reproduced: RMSE_η grows sub-linearly from ~ 0.02 m at step 10 (seed) to ~ 0.10 m at step 200, with no instability and $\leq 5\%$ visual deviation from the paper curve across all 54 cases. BEE is tightly clustered ($\text{IQR} \approx 0.024\text{--}0.040$), no outliers.

5.4 Reproduced figures and tables

Under `results/ja/`: `fig1_paper_aggregated.{pdf,png}` (paper Fig. 1), `fig5c_rmse_eta_vs_rollout_allcases.` (paper Fig. 5c), `fig6_runtime_summary.{pdf,png}` (paper Fig. 6), `table1_metrics_summary.csv` (paper Table 1/3 source-of-truth, 54 rows), and `table2_experimental_setup.csv` (this replication’s hyperparameter snapshot).

6 Verdict

REPLICATED. All four headline Table 3 metrics for the Selected / Test-EM cell are within the paper’s $1\text{-}\sigma$ band, with three of four agreeing to four decimal places and the fourth (ATE) better than the paper.

Self-score: Coverage 10/10 (all 54 cases, all four headline metrics, paper Figs. 1/5c/6 reproduced, Table S2 sampled). Agreement 10/10 (bullseye on RMSE/BEE; ATE delta favourable and explained).

7 Compute cost

Item	Value
GPU-hours	3.40 (single A100)
Wall-clock per case (mean $\pm$ std)	226.4 ± 2.8 s
Total wall-clock (54 cases, sequential)	3.40 h
Working storage on uicgpu	~ 150 GB
Dropbox payload (this report bundle)	~ 155 MB
Cash cost	\$0

The original paper’s training cost was on the order of tens of GPU-days. This replication paid roughly 1/5000 of that for inference.

8 Caveats

- Strict inference replication only — other variants (Baseline, ablations, training-from-scratch) are out of scope.
- Predicted peak η is on average slightly lower than truth (4.3 m vs. 5.6 m across 54 cases), a known F-FNO smoothing effect that the paper acknowledges and that does not affect the Table 3 metrics it claims.
- Per-case rollout CSVs other than the example case remain on uicgpu (/data/stevens/tsunami/results/Cas they are recoverable on demand but not mirrored to Dropbox to keep the payload reasonable.
- Inference is deterministic in eval mode; no random-seed control was needed.

9 Files

FFNO-Tsunami-Makarynskyy2026/

REPORT.md

report/ffno_tsunami_replication_report.{tex,pdf}

results/

metrics_summary.json

ja/ paper-figure-equivalents + tables

per_case_example/ worst-case full dump

Closeout completed 2026-05-27 by Ollie (OpenClaw subagent, model argo/argo:claude-opus-4.7).